Write a code using RAW sockets to implement packet sniffing

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Aim:

To write a program using RAW sockets to capture and display network packets passing through the network interface.

Algorithm:

→ Start the program.
→ Create a RAW socket using socket (AF_PACKET, SOCK_RAW, ...).
→ Bind the socket to the required network interface.
→ Continuously receive packets from the network.
→ Extract the Ethernet header from each packet.
→ Identify the protocol type, such as IPv4, ARP or IPv6
→ For IPv4 packets, extract the source and destination IP addresses.
→ Display the packet information.
→ Repeat until the user terminates the program.
→ Close the socket.

program - python:

```
import socket
import struct
# Create a RAW socket
raw_socket = socket.socket(
```

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```
        socket.AF_PACKET,
        socket.SOCK_RAW,
        socket.ntohs(3)
    )
print("packet Sniffer started...")
print("Waiting for packets...\n")
try:
  while True:
    # Receive packet
    packet, address = raw_socket.recvfrom(65535)
    # Ethernet header is first 14 bytes
    Ethernet_header = packet[:14]
    # Unpack Ethernet header
    dest_mac, src_mac, protocol = struct.unpack(
      "!6s6sH", ethernet_header)
    # Convert MAC addresses
    dest_mac = ":".join(
      "%02x"%b for byte in dest_mac
    )
    src_mac = ":".join(
      "%02x"%byte for byte in src_mac
    )
    print("source MAC :", src_mac)
    print("Destination MAC :", dest_mac)
    print("protocol :", hex(protocol))
    # IPv4 packet
    if protocol == 0x0800:
      ip_header = packet[14:34]
```

```
version_ihl, tos, total_length, identification, \ flags_
fragment, ttl, proto, checksum, \ src_ip, dest_ip
= struct.unpack("!BBHHHBBH4s4s", ip_header)
src_ip = socket.inet_ntoa(src_ip)
dest_ip = socket.inet_ntoa(dest_ip)
print("source Ip :", src_ip)
print("Destination Ip :", dest_ip)
print("TTL :", ttl)
print("Ip protocol :", proto)
elif protocol == 0x0806:
    print("packet Type : ARP")
elif protocol == 0x86DD:
    print("packet Type : Ipv6")
else
    print("packet Type : Other")
except keyboard Interrupt :
    print("\n packet sniffing stopped.")
finally :
    raw_socket.close()
```

~~Samp~~ Output :

```
packet Sniffer Started...
Waiting for packets...
-----------------------------------------
Source MAC : 3C:52:82:11:22:33
Destination MAC : ff:ff:ff:ff:ff:ff
protocol : 0x806
```

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packet Type : ARP

Source MAC : 3C:52:82:11:22:33

Destination MAC : 84:16:f9:44:55:66

protocol : 0x800

Source Ip : 192.168.1.10

Destination Ip : 192.168.1.1

TTL : 64

Ip protocol : 6

Source MAC : 84:16:f9:44:55:66

Destination MAC : 3C:52:82:11:22:33

protocol : 0x800

Source Ip : 192.168.1.1

Destination Ip : 192.168.2.10

TTL : 64

Ip protocol : 6

Result :

Thus, a packet-sniffing program using RAW sockets was successfully implemented. The program captures network packets and displays important information such as source MAC address, destination MAC address, protocol, source Ip address, destination Ip address and TTL.

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